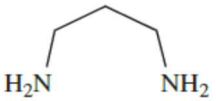
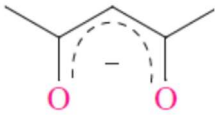
Q1. Draw the structure of ligand and predict the denticity:

a. Ethylene diamine
b. Propylene diamine
c. Acetylacetonate ion
d. Oxalate ion
e. 2, 2' bipyridine
f. 1, 10 phenanthroline
g. $[EDTA]^{4-}$

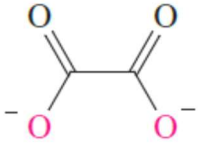
1,2-ethanediamine



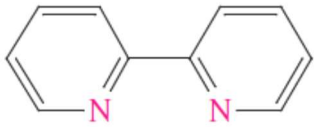
1,3-propanediamine

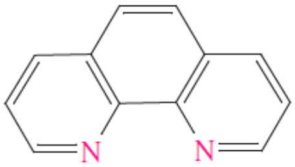


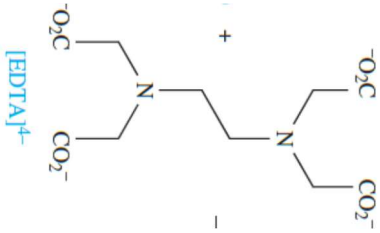
Acetylacetonate ion



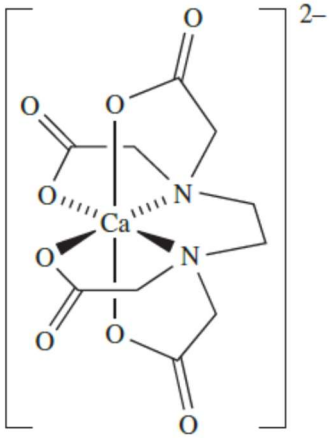
Oxalate ion

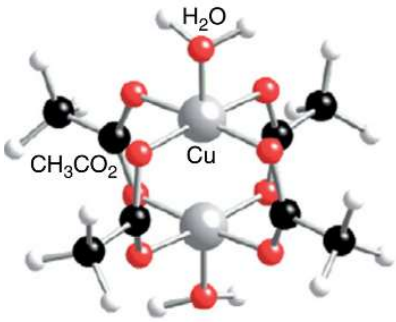




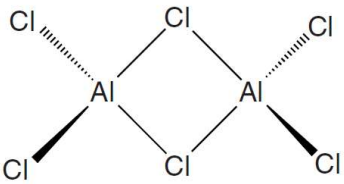


Q2. Draw the structure of $[Ca(EDTA)]^{2-}$, $Cu(CH3COO)_2.H_2O$ structure

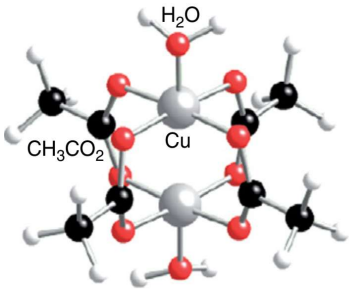




Q3. Write the correct formula of the complex using the bridging convention



$[Al_2Cl_4(\mu\text{-}Cl)_2]$ or $[Cl_2Al(\mu\text{-}Cl)_2AlCl_2]$



$[(H_2O)Cu(\mu\text{-}CH_3CO_2)_4Cu(OH_2)]$

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Tutorial 2 (CH1101, Module 2)

Question 1. Draw the isomers of the following complexes and name the isomerism.

(i) tetraammineaquachlorocobalt(III) chloride;

(ii) pentaammineaquacobalt(III) nitrate;

(iii) tetraamminechloronitritocobalt(III) chloride;

(iv) tris(ethylenediamine)cobalt(III) hexacyanochromate(III);

(v) tetraammineplatinum(III) hexachloroplatinate(III);

(vi) pentaamminenitritocobalt(III) chloride

Answer: (i) $[Co(NH_3)_4(H_2O)Cl]Cl_2$ and $[Co(NH_3)_4Cl_2]Cl{\cdot}H_2O$ – Hydrate isomer

(ii) $[Co(NH_3)_5(H_2O)](NO_3)_3$ and $[Co(NH_3)_5(NO_3)](NO_3)_2{\cdot}H_2O$ – Hydrate isomer

(iii) $[Co(NH_3)_4(NO_2)Cl]Cl$ and $[Co(NH_3)_4Cl_2]NO_2$ – Ionization isomer

(iv) $[Co(en)_3][Cr(CN)_6]$ and $[Cr(en)_3][Co(CN)_6]$ – Coordination isomer

(v) $[Pt(NH_3)_4][PtCl_6]$ and $[PtCl_2(NH_3)_4][PtCl_4]$ – Coordination isomer

(vi) $[Co(NO_2)(NH_3)_5]Cl_2$ and $[Co(ONO)(NH_3)_5]Cl_2$ – Linkage isomer

Question 2. What chemical tests would you use to distinguish between:

(i) $[Co(NH_3)Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$,

(ii) $[CrCl_2(H_2O)_4]Cl.H_2O$ and $[CrCl(H_2O)_5]Cl_2$?

Answer:

$[Co(NH_3)Br]SO_4 \xrightarrow{Ba^{2+}} BaSO_4$ precipitate

$[Co(NH_3)_5SO_4]Br \xrightarrow{Ba^{2+}}$ No reaction

$[CrCl_2(H_2O)_4]Cl.H_2O \xrightarrow{Ag^+} AgCl$

$[CrCl(H_2O)_5]Cl_2 \xrightarrow{Ag^+} 2\ AgCl$

Question 3. Draw the possible isomers of the complexes (i) $[PtCl_2(PMe_3)_2]$; (ii) $[PtBr_2(py)_2]$; (iii) $[SnF_4(Me)_2]^{2-}$;

(iv) $[CoCl_3F_3]$

Answer:

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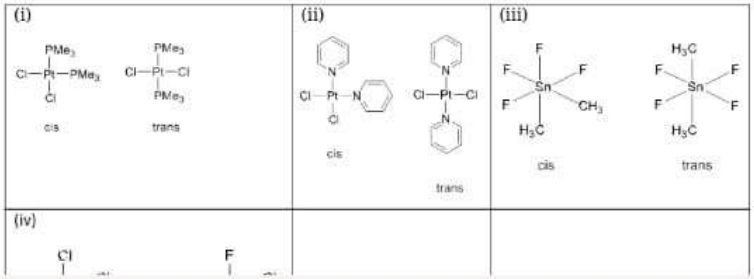

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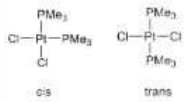
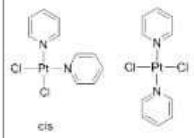
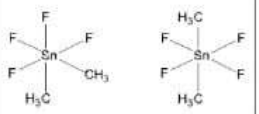
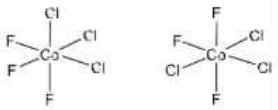
Question 3. Draw the possible isomers of the complexes (i) $[PtCl_2(PMe_3)_2]$; (ii) $[PtBr_2(py)_2]$; (iii) $[SnF_4(Me)_2]^{2-}$;

(iv) $[CoCl_3F_3]$

Answer:

Tutorial 2 (CH1101, Module 2)

(i)  cis trans	(ii)  cis trans	(iii)  cis trans
(iv)  fac mer		

Question 4. Which of the following octahedral complexes are chiral: cis-$[CoCl_2(en)_2]^+$, $[Cr(ox)_3]^{3-}$, trans- $[PtCl_2(en)_2]^{2+}$, $[Ni(phen)_3]^{2+}$?

Answer: cis-$[CoCl_2(en)_2]^+$; $[Cr(ox)_3]^{3-}$; $[Ni(phen)_3]^{2+}$ **[Explain by drawing structure]**

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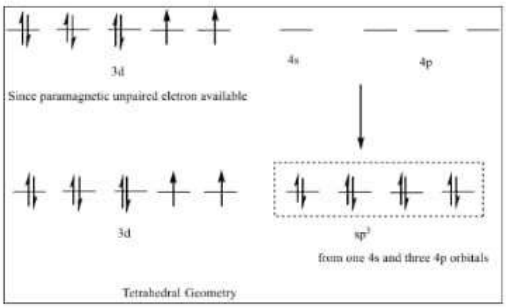
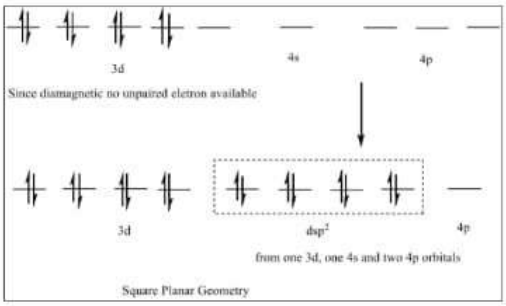
Tutorial 3 (CH1101, Module 2)

Question 1. Using valence bond theory, predict the hybridization and geometry of paramagnetic $[NiCl_4]^{2-}$, diamagnetic $[Ni(CN)_4]^{2-}$, diamagnetic $[Co(NH_3)_6]^{3+}$, paramagnetic $[CoF_6]^{3-}$, and paramagnetic $[FeBr_4]^{2-}$

Answer: (i) $[NiCl_4]^{2-}$

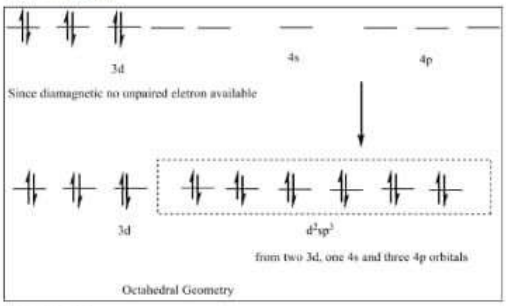


(ii) $[Ni(CN)_4]^{2-}$



Tutorial 3 (CH1101, Module 2)

(iii) $[Co(NH_3)_6]^{3+}$



(iv) $[CoF_6]^{3-}$

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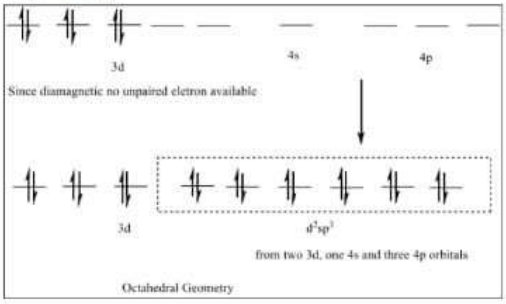
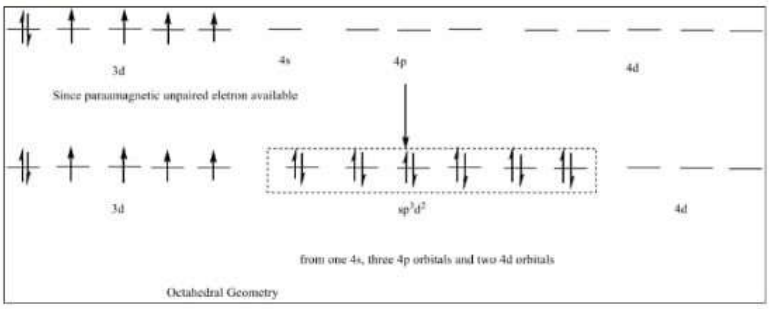
Tutorial 3 (CH1101, Module 2)

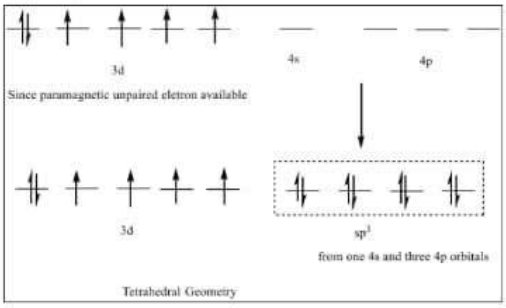
(iii) $[Co(NH_3)_6]^{3+}$



(iv) $[CoF_6]^{3-}$



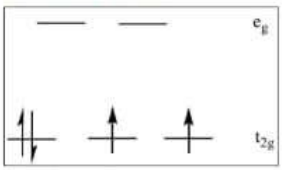
(v) $[FeBr_4]^{2-}$



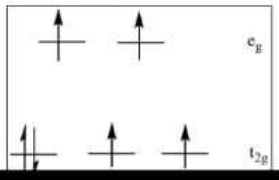
Tutorial 3 (CH1101, Module 2)

Question 2. Draw the crystal field diagrams of $[Mn(CN)_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

Answer: (i) $[Mn(CN)_6]^{3-}$



(ii) $[Fe(H_2O)_6]^{2+}$



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Tutorial 3 (CH1101, Module 2)

Question 1. Using valence bond theory, predict the hybridization and geometry of paramagnetic $[NiCl_4]^{2-}$, diamagnetic $[Ni(CN)_4]^{2-}$, diamagnetic $[Co(NH_3)_6]^{3+}$, paramagnetic $[CoF_6]^{3-}$, and paramagnetic $[FeBr_4]^{2-}$

Question 2. Draw the crystal field diagrams of $[Mn(CN)_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

Question 3. Calculate the CFSE of $[Mn(H_2O)_6]^{2+}$, $[Co(H_2O)_6]^{2+}$, and $[Co(CN)_6]^{4-}$

Question 4. Determine the number of unpaired electrons in octahedral complexes of d^9, d^8, d^5, d^4 and d^3 ions when $\Delta_o > P$ and $\Delta_o < P$

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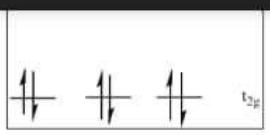

Question 3. Calculate the CFSE of $[Mn(H_2O)_6]^{2+}$, $[Co(H_2O)_6]^{2+}$, and $[Co(CN)_6]^{4-}$

Answer:

(i) $[Mn(H_2O)_6]^{2+}$,

Mn^{2+} is d^5, so $t_{2g}^3\, e_g^2$

CFSE = 3 x 0.4 Δ_o – 2 x 0.6 Δ_o = 0

(ii) $[Co(H_2O)_6]^{2+}$,

Co^{2+} is d^7, so $t_{2g}^5\, e_g^2$

CFSE = 5 x 0.4 Δ_o – 2 x 0.6 Δ_o = 0.8 Δ_o

(iii) $[Co(CN)_6]^{4-}$,

Co^{2+} is d^7, but low spin, so $t_{2g}^6\, e_g^1$

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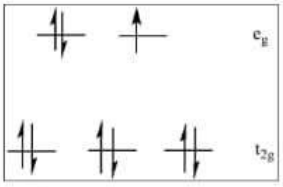
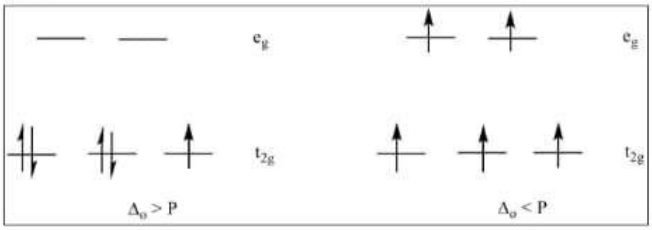
Tutorial 3 (CH1101, Module 2)

Question 4. Determine the number of unpaired electrons in octahedral complexes of d^9, d^8, d^5, d^4 and d^3 ions when $\Delta_o > P$ and $\Delta_o < P$

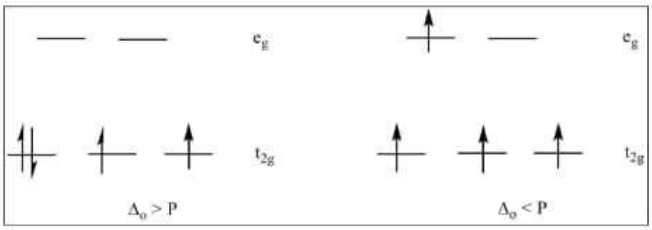
Answer: (i) d^9 - for both $\Delta_o > P$ and $\Delta_o < P$ one unpaired electron



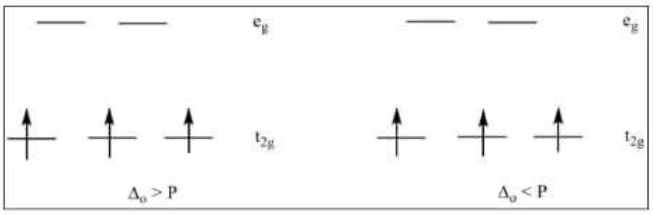
(ii) d^5 –



(iii) d^4 –



(iv) d^3 –



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Tutorial 3 (CH1101, Module 2)

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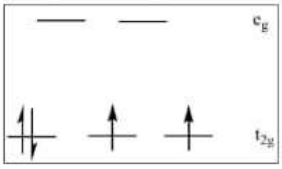

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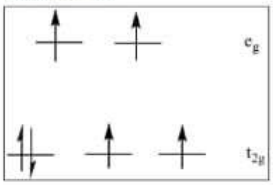
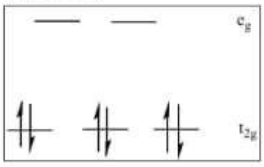


(iii) $[Fe(CN)_6]^{4-}$



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CFSE = 3 x 0.4 Δ_o – 2 x 0.6 Δ_o = 0

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Tutorial 3 (CH1101, Module 2)

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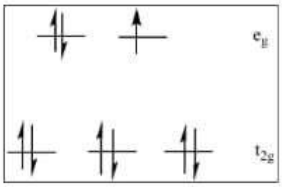

(ii) d^5 –